

³³P - Comments on evaluation of the decay data

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This evaluation has been carried out with the literature available by March 2026. The former evaluation was completed in 2004 (2004BEZR). Datasets have been averaged with the LWIGHT (*Limitation of Relative Statistical Weight*) program.

1. Nuclear data

Phosphorus-33 disintegrates by β^- emission (100%) to the ground state of Sulfur-33, as the first excited state of ³³S is above the Q-value of ³³P. The Q-value (248.5 (11) keV) has been adopted from the latest Atomic Mass Evaluation AME2020 (2021WA16). The spins and parities of the ground states of the two nuclei have been adopted from (1967EN05): 1/2⁺ for ³³P and 3/2⁺ for ³³S.

1.1. Half-life

The experimental values (in days) are given in Table 1 that contains 9 initial values. From these, the first one is without uncertainties (1951SH92) and consequently is omitted from the analysis. Two other series come from the same laboratories, so only the last value is retained for each of them: 1954NI06 updated 1952Je12 and 1960FI05 superseded 1952WE29 and 1954WEST.

The initial average calculation on the 5 remaining independent data using LWIGHT resulted in a reduced- χ^2 of 8.07, to be compared to a critical χ^2 of 3.32. The dataset is discrepant due to the value of 1954NI06 which is significantly lower than the other values. The announced uncertainty of 0.2 d has been increased to 0.5 d, which resulted in a consistent dataset, with a reduced- χ^2 of 3.32, equal to the critical one.

The final calculated result is 25.380 (40) d with an internal uncertainty of 0.04 d, an external uncertainty of 0.07 d. This weighted average is mainly influenced by the two last measurements (1968RE04 weighs 65%, while 1972LA14 weighs 33%), which are not entirely consistent with one another.

Therefore, the adopted value is 25.38 (5) d, keeping the final uncertainty at the same level as the minimum experimental uncertainty.

Table 1 – ³³P half-life experimental values.

Reference	T _{1/2} (d)	Comment
1951SH92	25	Excluded, no uncertainty
1952JE12	24.8 (5)	Superseded by 1954NI06
1952WE29	25 (2)	Superseded by 1954WEST and 1960FI05
1954NI06	24.4 (5)	Published uncertainty (0.2) increased
1954WEST	25.4 (2)	Superseded by 1960FI05
1958RU66	25 (1)	
1960FI05	25.2 (5)	
1968RE04	25.30 (5)	
1972LA14	25.56 (7)	
Adopted	25.38 (5)	Weighted average with minimum experimental uncertainty

1.2. Beta transition

There is only a single β^- transition in the ³³P decay scheme that directly populates the ground state of ³³S; this ground-state-to-ground-state transition is of allowed nature. The transition energy corresponds to the adopted Q_β = 248.5 (11) keV.

Table 2 – Experimental values of the end-point of the β^- spectrum of ³³P.

Reference	E _{max} (keV)	Comment
1951SH92	270 (20)	Excluded by the Chauvenet criterion
1952JE12	260 (20)	Superseded by 1954NI06
1952WE29	246 (5)	Superseded by 1954WEST
1954NI06	249.0 (20)	
1954EL25	251 (5)	
1954WEST	246 (5)	
1958RU66	238 (5)	Excluded by the Chauvenet criterion
1984PO09	248.3 (13)	
Calculated	248.5 (10)	Weighted average

The eight experimental values of the maximum energy of the β^- continuous spectrum are presented in Table 2. Two of them were updated following new measurements at the same laboratory: 1952JE12 and 1952WE29 were updated by 1954NI06 and 1954WEST, respectively; only the last value is retained for each of them.

The weighted mean of the β^- end-point energy has been calculated from the six remaining values, using LWHEIGHT. According to the averaging procedure, 1951SH92 and 1958RU66 are excluded by the Chauvenet criterion.

The averaging resulted in a reduced χ^2 of 0.20, compared with a critical χ^2 of 3.79, with an internal uncertainty of 1 keV, an external uncertainty of 0.5 keV. **The calculated β^- end-point energy is 248.5 (10) keV**, consistent with the adopted Q-value.

The $\log ft = 5.033$ (7) value have been calculated by means of the BetaShape code version 2.4 (2023MO21). This program also provides an **average spectrum energy of 75.91 (38) keV** from theory, which has been adopted.

2. Atomic data

Fluorescence yield data, relative K X-ray emission probabilities and ratios $P(KLX)/P(KLL)$ and $P(KXY)/P(KLL)$ are from Schönfeld *et al.* (1996SC06). Table 3 summarizes the X-ray emission energies and relative intensities of K X rays of the daughter element (S).

Table 3 – X-ray emission energies of sulfur.

Origin	Energy (keV)	Relative probability (%)
XL (S)	0.1513 – 0.2272	
XK _{α2} (S)	2.30667	50.53
XK _{α1} (S)	2.30787	100
XK _{β1} (S)	2.4641	9.33

3. Production modes

- $^{30}\text{Si} (\alpha, p) ^{33}\text{P}$
 - $^{33}\text{S} (n, p) ^{33}\text{P}$
 - $^{34}\text{S} (n, d) ^{33}\text{P}$
 - $^{36}\text{Cl} (n, \alpha) ^{33}\text{P}$
- Possible impurities: ^{32}P (fast neutrons), ^{35}S (thermal neutrons)

4. Recommendation of measurements

The amount of data available to characterize ^{33}P decay data is extremely limited.

5. References

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1952WE29	Westermarck, Torbjörn. «Production of P^{33} with Thermal Neutrons.» <i>Physical Review</i> 88 (November 1952): 573–574. [Half-life, $E_{\max}$]
1954NI06	Nichols, R. T. and E. N. Jensen. «Radioisotope P^{33} .» <i>Physical Review</i> 94 (April 1954): 369–371. [Half-life, $E_{\max}$]
1954EL25	Elbek, B., K. O. Nielsen and O. B. Nielsen. «Determination of Maximum Beta Energy in the Decay of P^{33} .» <i>Physical Review</i> 95 (July 1954): 96–97. [$E_{\max}$]
1954WEST	Westermarck, T. «Production and properties of P^{33} .» <i>Arkiv för Fysik</i> 7 (1954): 87. [Half-life, $E_{\max}$]
1958RU66	Russell, J. E. «Reactions for the production of P^{33} .» <i>Nuclear Science and Engineering</i> , 1958. [Half-life, $E_{\max}$]
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1984PO09	Polak, P. and L. Linder. «Liquid Scintillation Beta Spectroscopy on ^{32}Si .» <i>Radiochimica Acta</i> 35 (1984): 23–27. [$E_{\max}$]
1996SC06	Schönfeld, E. and H. Janßen. «Evaluation of atomic shell data.» <i>Nuclear Instruments and Methods in Physics Research Section A</i> : 369 (February 1996): 527–533. [Atomic data]
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